

Quantifying Effect Modifier Similarity for Regional Pooling in Multi-Regional Clinical Trials

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1. Motivation

Multi-regional clinical trials (MRCTs) are the standard paradigm for global drug development. ICH E17 describes pooling decisions in terms of *similarity of effect modifier distributions* but gives **no quantitative metric, threshold, or procedure**.

Why it matters. With CATE $\tau(x) = E[Y(1) - Y(0) | X = x]$,

$$\hat{\tau}_r = \int \tau(x) dF_r(x).$$

so regions with different effect modifier distributions can observe different treatment effects even when the drug works identically at the individual level.

Gap. The standardized mean difference (SMD) only detects *location* shifts. $N(50, 5^2)$ vs. $N(50, 15^2)$ yields SMD= 0 despite a 3-fold difference in spread; scale and shape differences are invisible.

Objective. A distributional dissimilarity index that (i) captures features beyond location, (ii) has a clinically interpretable scale, and (iii) carries a theoretical link to treatment-effect heterogeneity.

2. The nABCD Dissimilarity Index

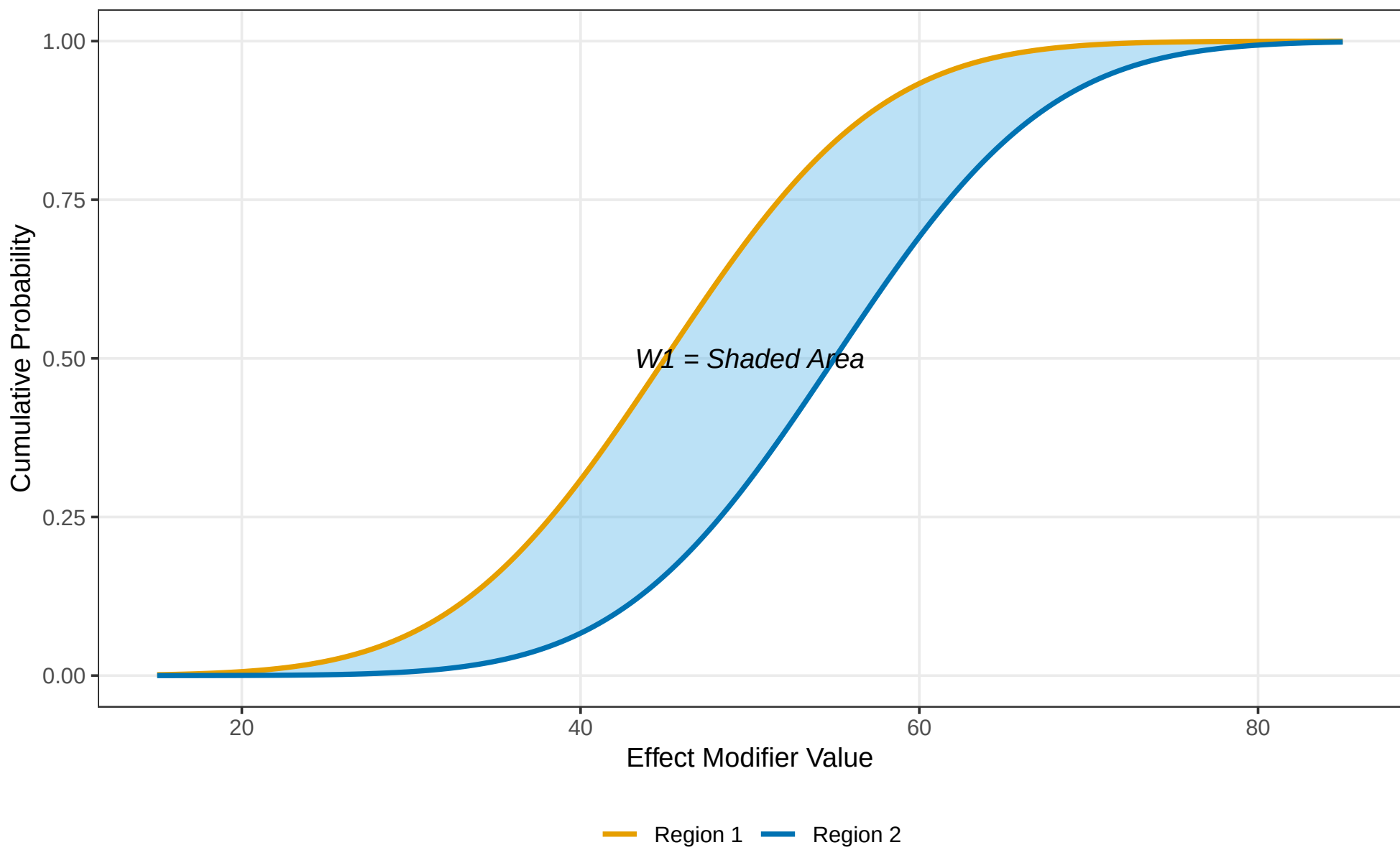
Wasserstein-1 distance between CDFs:

$$W_1(F, G) = \int_{-\infty}^{\infty} |F(x) - G(x)| dx,$$

normalized by a robust scale:

$$nABCD(F_1, F_2) = \frac{W_1(F_1, F_2)}{2 \cdot IQR_{pooled}}$$

where IQR_{pooled} is computed from $F_{pooled} = \lambda F_1 + (1 - \lambda) F_2$, $\lambda = n_1/(n_1 + n_2)$. The factor of 2 calibrates so that a 1-IQR location shift yields nABCD = 0.5.



Properties.

- ≥ 0 ; $= 0 \iff F_1 = F_2$; symmetric; scale-free.
- Responds to location, scale, skewness, and shape.
- Requires only *baseline* effect modifier data – from prior trials, registries, or RWE.

3. Heterogeneity Bound & Clinical Calibration

By Kantorovich–Rubinstein duality,

$$W_1(F_1, F_2) = \sup_{\|f\|_{Lip} \leq 1} \left| \int f dF_1 - \int f dF_2 \right|.$$

If $\tau(x)$ is Lipschitz with constant L , then τ/L is 1-Lipschitz and

$$|\hat{\tau}_1 - \hat{\tau}_2| \leq L \cdot W_1(F_1, F_2).$$

Substituting gives the **clinical heterogeneity bound**:

$$\hat{\tau}_{max} = 2L \cdot IQR_{pooled} \cdot nABCD(F_1, F_2)$$

with bootstrap CI by linear mapping of the nABCD percentile interval. L is the per-unit CATE slope from Phase 2 interaction analysis or subgroup-specific meta-analyses.

A W_1 -specific property: W_2 and KS admit no analogous heterogeneity bound (Gibbs & Su, 2002).

4. L^* Reverse Calculation (unknown L)

At the planning stage, L is rarely numerically available. Rearranging the heterogeneity bound:

$$L^* = \frac{\hat{\tau}_{min}}{2 \cdot IQR_{pooled} \cdot nABCD}$$

L^* is the **CATE sensitivity required** for the observed distributional difference to yield a clinically meaningful $\hat{\tau}_{min}$.

Decision workflow.

- Rank candidate partners by $nABCD$ with bootstrap CIs.
 - Report L^* at pre-specified $\hat{\tau}_{min}$.
 - Sponsor judges plausibility of L^* for the effect modifier and drug class.
- This *estimation-centered* framework deliberately avoids binary accept/reject testing and fixed nABCD cutoffs.

5. Implementation & Data Requirements

Inference. Percentile bootstrap ($B \geq 2,000$) for $nABCD$ CIs; L^* CI obtained by the same linear percentile mapping. BCa showed inferior coverage for this bounded statistic. Recommended $n \geq 100$ per region.

Data source flexibility. nABCD requires only *baseline* effect modifier distributions – not treatment outcomes. Regional data may be drawn from prior trials, disease registries, electronic health records, or other real-world evidence (RWE) sources, aligning with ICH E6(R3). Representativeness of each region's data with respect to the target trial population is the sole prerequisite.

Bridge to validation. The next column quantifies how reliably $nABCD$ recovers the true index and how it compares with SMD across systematic distributional alterations.

Software. Reference R implementation available; an R package with bootstrap inference and clinical calibration is under development.

6. Simulation Design (8 Scenarios)

Systematic scenarios isolating location, scale, shape, and skew. 10,000 replications each; $n \in \{50, 100, 200\}$ per region; $B = 2,000$ bootstrap resamples.

ID	Type	F_1 vs. F_2	True
S1	Null	$N(50, 10^2)$ vs. same	0.000
S2	Loc. 0.2σ	vs. $N(52, 10^2)$	0.075
S3	Loc. 0.5σ	vs. $N(55, 10^2)$	0.180
S4	Loc. 1.0σ	vs. $N(60, 10^2)$	0.328
S5	Scale 1.5×	vs. $N(50, 15^2)$	0.122
S6	Shape (Gamma)	vs. $\text{Gamma}(25, 0.5)$	0.024
S7	Skew (LogN)	vs. $\text{LogN}(\mu, \sigma)$	0.304
S8	Loc + Scale	vs. $N(55, 15^2)$	0.175

S8 is the most realistic MRCT pattern; S7 matches biomarker CVs= 53%; S6 isolates shape alone.

7. Estimation Quality: Bias & Coverage

Bias = Mean($nABCD$) – $nABCD$ and coverage of 95% percentile bootstrap CIs at $n = 100$ per region:

Scenario	True	Bias	Coverage
S1 (Null)	0.000	0.066	–1
S2 (0.2σ)	0.073	0.020	0.981
S3 (0.5σ)	0.180	0.003	0.991
S4 (1.0σ)	0.328	0.003	0.990
S5 (Scale)	0.122	0.014	0.910
S6 (Shape)	0.024	0.046	
S7 (Skew)	0.304	0.008	0.984
S8 (Loc+Scale)	0.175	0.011	0.935

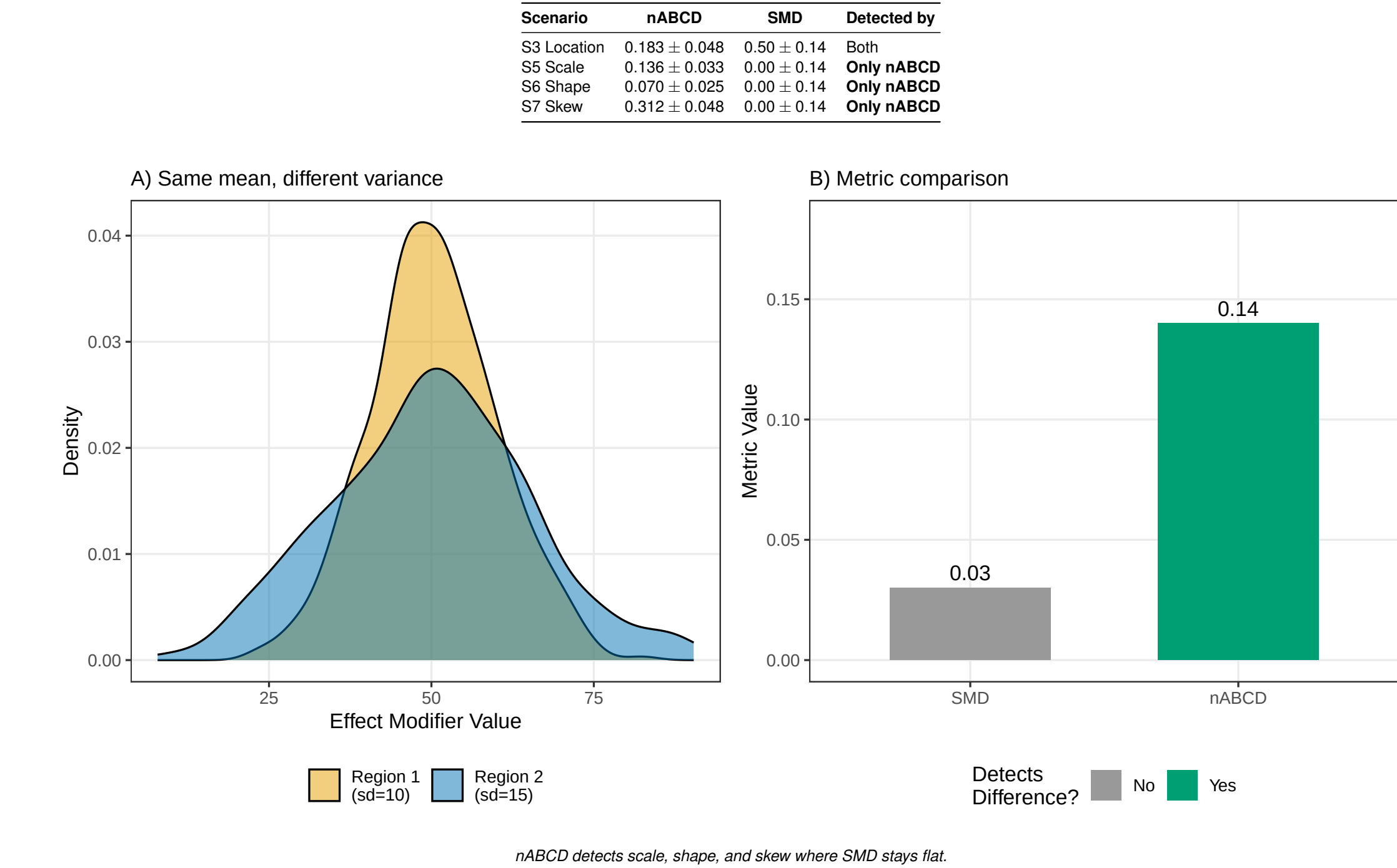
¹Near-boundary true value; non-negative constraint induces positive bias that prevents meaningful coverage at these sample sizes (Andrews, 2000).

Take-aways.

- True $nABCD \geq 0.1$: bias < 0.02, coverage 0.88–0.96 at $n \geq 100$.
- **Percentile** bootstrap preferred over BCa (boundary).
- Recommended: $n \geq 100$ per region.

8. nABCD vs. SMD: Beyond Location

At $n = 100$ per region, SMD is blind to precisely the features that can drive non-linear effect-modifier heterogeneity:



nABCD detects scale, shape, and skew where SMD stays flat.

9. Framework Philosophy

- **Estimation-centered, not testing.** Report $nABCD$, bootstrap CI, and $\hat{\tau}_{min}$ or L^* – never a binary verdict.
- **No universal cutoff.** Cohen's d -style benchmarks are context-free; nABCD's clinical weight depends on L , which is external to the distributional comparison.
- **Ranking + calibration.** Rank partners by $nABCD$; translate to clinical scale via $\hat{\tau}_{min}$ (L known) or L^* (L unknown).
- **Two-pathway calibration.** Same framework for confirmatory (Phase 2 L) and planning (sensitivity via L^*) settings.
- **ICH E17 aligned.** "Similar enough" remains with sponsor and regulators, informed by transparent quantitative inputs.

Limitations. Continuous effect modifiers only; univariate (multivariate OT extension is future work); near-boundary positive bias for true $nABCD \lesssim 0.05$; requires representative baseline data.

10. Case Study: Hypothetical Drug T MRCT

A sponsor develops **Drug T**, a novel thrombolytic for acute myocardial infarction, and plans a Phase 3 MRCT. At the planning stage the sponsor uses IPD from **GUSTO-4** ($N = 40,830$; 16 anonymized regions) to assess distributional similarity of candidate effect modifiers across regions and to identify pooling partners for a designated anchor region.

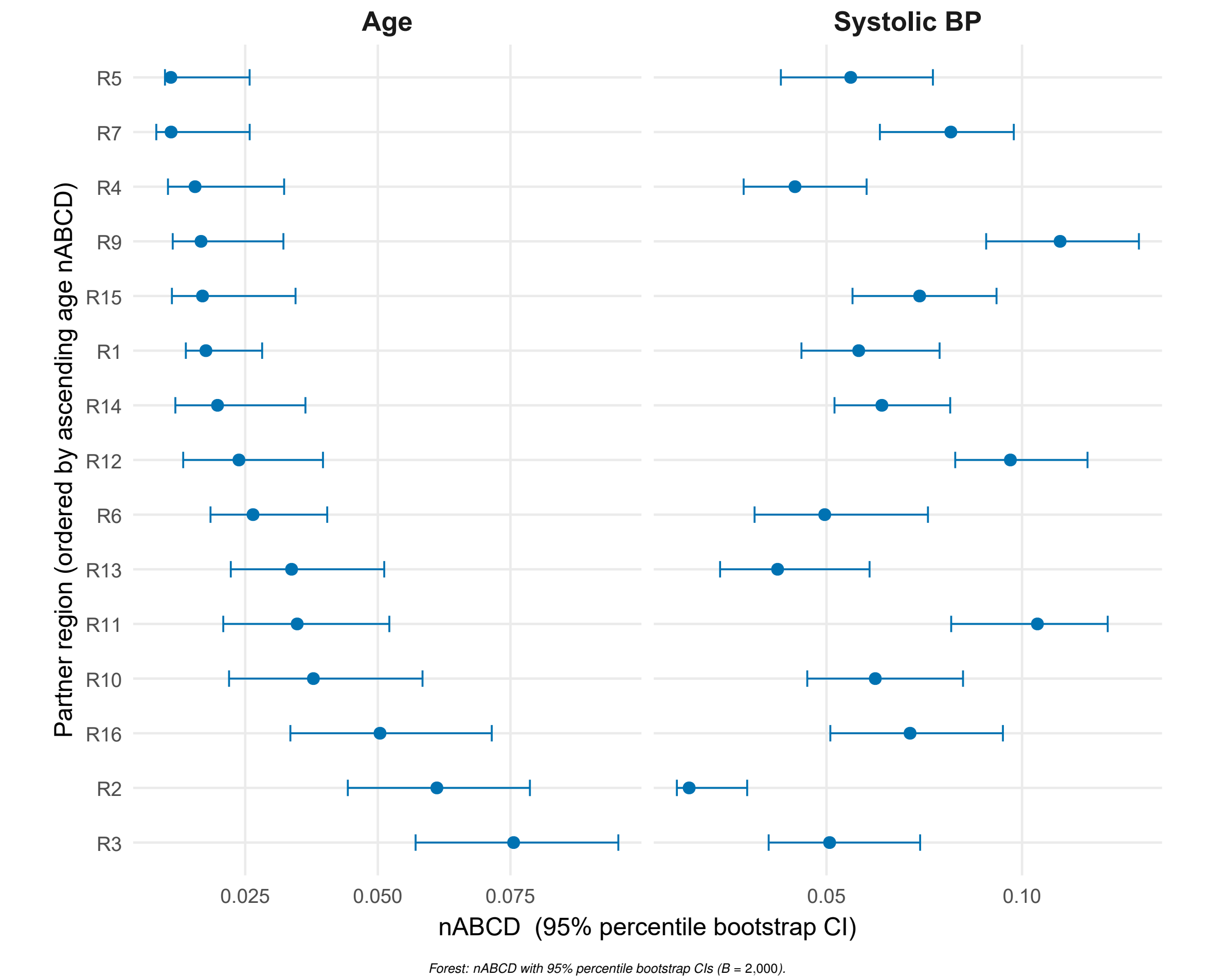
Candidate effect modifiers. Age and systolic blood pressure (SBP), flagged by Drug T's Phase 2 exploratory analyses; class-level numerical CATE slopes are unavailable (FTT meta-analysis reports benefit "irrespective of age" without per-year slope). L is therefore treated as unknown and the L^* pathway applies to both.

Region 8 as anchor ($n = 2,916$).	Candidate	EM	Mean	SD	Skew	IQR_{pooled}
	Age (yr)		60.2	12.1	–0.17	17–18
	SBP (mmHg)		132.4	22.9	0.20	30–34

GUSTO-4 was not designed as an MRCT; regions are anonymized, which precludes demographic or geographic interpretation. The dataset serves here purely as a publicly available vehicle for methodological demonstration.

11. nABCD Across 15 Partner Regions

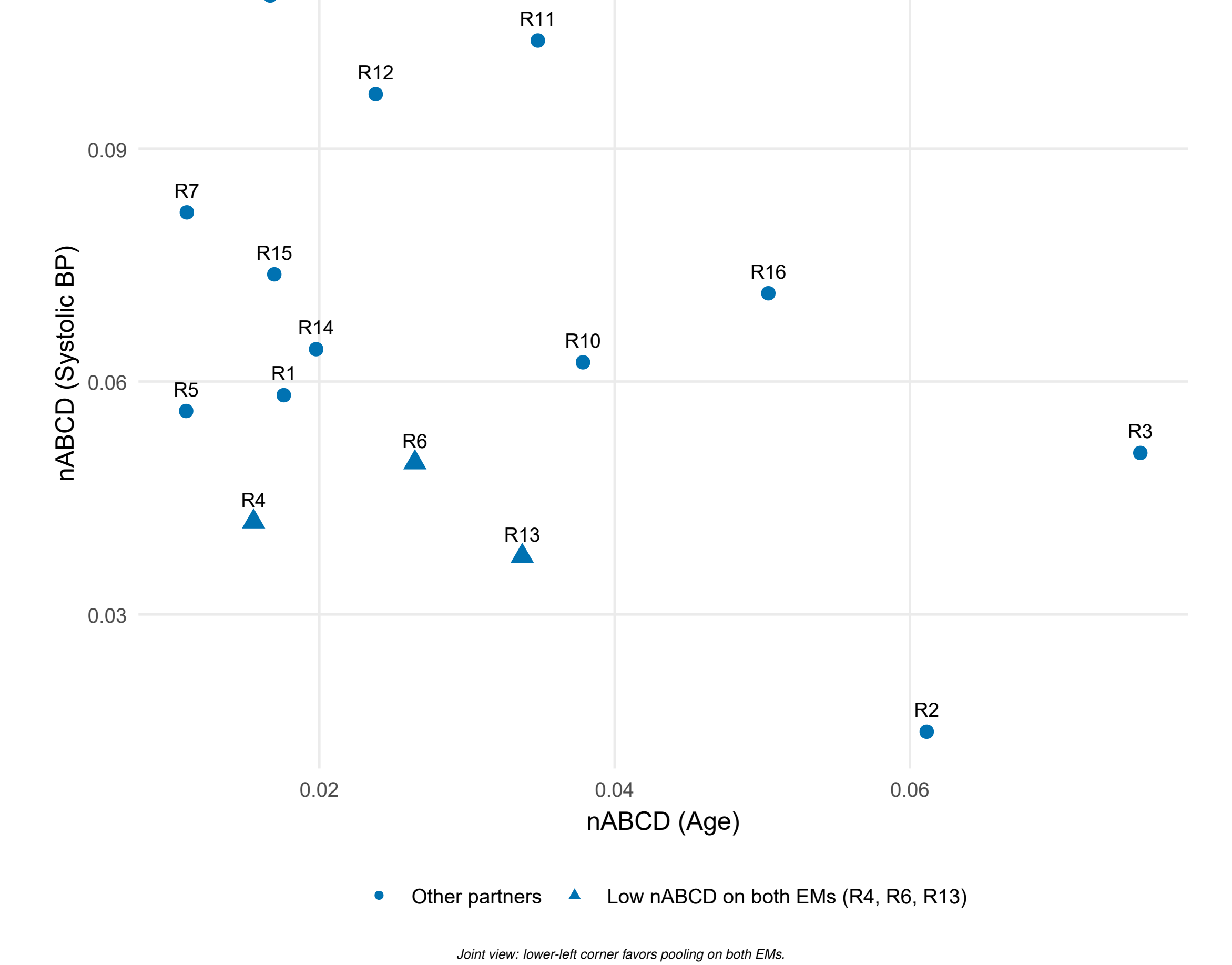
Rank	Partner	n	$nABCD_{age}$ [95% CI]	$nABCD_{sbp}$ [95% CI]
1	R5	1909	0.011 [0.010, 0.020]	0.056 [0.038, 0.077]
2	R7	3150	0.011 [0.008, 0.026]	0.082 [0.064, 0.098]
3	R4	2876	0.016 [0.010, 0.032]	0.042 [0.020, 0.060]
4	R9	3123	0.017 [0.011, 0.032]	0.110 [0.091, 0.130]
5	R15	2576	0.017 [0.011, 0.034]	0.074 [0.057, 0.093]
14	R2	2852	0.081 [0.044, 0.079]	0.015 [0.012, 0.030]
15	R3	2020	0.076 [0.057, 0.095]	0.051 [0.035, 0.074]



Forest: nABCD with 95% percentile bootstrap CIs ($B = 2,000$).

Age vs. SBP patterns differ. Age nABCD spans 0.011–0.076 (11/15 partners < 0.04); SBP nABCD spans 0.015–0.110 with most partners in 0.050–0.110 – SBP distributions are more heterogeneous across GUSTO-4.

R2 vs. R9 asymmetry. R2 has the smallest SBP nABCD (0.015) but the 2nd-largest age nABCD (0.061); R9 shows the opposite (age 0.017; SBP 0.110). A single-modifier ranking would draw opposite conclusions.



12. Clinical Interpretation via L^*

L^* at $\hat{\tau}_{min} = 1\%$ pt (absolute 30-day mortality), selected rows:

Partner	$nABCD_{age}$	$nABCD_{sbp}$	L^* age (yr)	L^* SBP (mmHg)
R2	0.061	0.015	0.0047	0.0105
R4	0.016	0.042	0.0180	0.0038
R6	0.026	0.030	0.0110	0.0030
R9	0.017	0.110	0.0171	0.0015
R13	0.034	0.037	0.0087	0.0042

